



Exposing fake images generated by text-to-image diffusion models

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ABSTRACT

Text-to-image diffusion models (DM) have posed unprecedented challenges to the authenticity and integrity of digital images, which makes the detection of computer-generated images one of the most important image forensics techniques. However, the detection of images generated by text-to-image diffusion models is rarely reported in the literature. To tackle this issue, we first analyze the acquisition process of DM images. Then, we construct a hybrid neural network based on attention-guided feature extraction (AGFE) and vision transformers (ViTs)-based feature extraction (ViTFE) modules. An attention mechanism is adopted in the AGFE module to capture long-range feature interactions and boost the representation capability. ViTFE module containing sequential MobileNetv2 block (MNV2) and MobileViT blocks are designed to learn global representations. By conducting extensive experiments on different types of generated images, the results demonstrate the effectiveness and robustness of our method in exposing fake images generated by text-to-image diffusion models.

1. Introduction

With the recent release of several generative models, including Dall-E 2 [1], Imagen [2], and Stable Diffusion [3], diffusion models (DM) has become a popular choice for digital content generation. Designing an automatic system that generates visually realistic images from textual descriptions is a non-trivial task and therefore can be seen as a major milestone toward human-like or general artificial intelligence [4]. However, the abuse of image generation tools makes it easy to generate fake images to mislead the public, which makes it increasingly urgent to ensure the integrity and authenticity of images [5–7].

To differentiate fake images from real ones, active and passive forensics approaches have been proposed in recent years. Active forensics methods typically involve designing various watermarks or fingerprints and embedding them into digital images [8,9]. By verifying the presence of such embedded information, the authentication of the images can be verified [10]. Although the methods can provide valid identification results, they require implementing the information embedding operation on the image capture device, which limits the application in real-world scenarios. Therefore, passive (blind) forensics methods have attracted an increasing amount of attention, which only rely on the inherent trace of digital images for detection [9].

For the passive forensics of digital images, most algorithms are designed to distinguishing between natural photographic (NP) and traditional computer-generated (TCG) images or generative adversarial network (GAN) images. For example, in [11], Ng et al. found the gamma correction and the sharp structures in CG and NP images are different, and the authors designed object geometry features for the detection of CG images. Wu et al. [12] designed a method by utilizing several visual features derived from color, edge, saturation, and texture. Experimental results demonstrate that the proposed features are effective under specific scenarios. In [13], Dehnie et al. introduced a wavelet-based denoising filter to extract the residual image, which can be used for detection. In [14], the authors pointed out that the photo-realistic TCG images are more surrealist and smoother than natural images, they leveraged image perception characteristics to detect TCG images. Since texture feature plays an important role in computer vision, more attention has focused on the usage of texture-based features for automatic TCG and NP classification. For example, Li et al. [15] utilized uniform gray-scale invariant local binary patterns to distinguish TCG images from NP images. The authors in [16] combined 31-dimensional statistical and textural features to discriminate the acquisition pipelines of digital images. They also constructed a 9-D

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histogram feature and a 9-D multi-fractal spectrum feature to represent the distinct texture. In addition, the distribution of histogram [17], spatial-temporal variation [18], physiological signal [19], quaternion wavelet transform features [20] and visual features [21] are also used to detect TCG images.

Inspired by the recent success of deep neural networks, many deep learning-based methods have also been proposed to distinguish CG images from NP images. For example, in [22], Rahmouni et al. designed a patch-based Convolutional neural network (CNN) with a custom pooling layer to detect TCG images. In [23], the authors constructed a convolution neural network trained on image patches, and the detection accuracy reached 98.5% in their scenario. In [24], Quan et al. introduced a generic framework that uses CNN to identify NP and TCG images. The network can be adjusted to handle different sizes of input image patches. Since Sensor pattern noise (SPN) is an effective and robust device fingerprint that can be used for a variety of important digital image forensic tasks, Yao et al. proposed a method based on SPN and deep learning to distinguish CGs from NP images [25]. The results demonstrated that the model with three high-pass filters can outperform that with only one or no filter on TCG image detection. Zhang et al. proposed a CNN-based model with a self-coding module to extract the correlation between color channels for the task [26]. Many researchers also designed neural networks with multi-branches to improve the performance. For example, in [27], the authors constructed a CNN with two-stream, one of which is designed for trace learning using a pre-trained VGG-19 network, and the other stream extracts noise-based distinct features to enhance performance. In [28], He et al. proposed an attention-based dual-branch convolutional neural network (AD-CNN) to extract robust representations from fused color components.

A series of methods adopting deep neural networks were also proposed to identify NP and GAN images. Mo et al. analyzed the difference of high-frequency components and proposed a CNN-based method for GAN image detection [29]. In [30], Li et al. leveraged representation learning and representation comparison to enhance the performance of GAN image detection. In [31], Fu et al. designed a dual-channel CNN to extract robust representations for the detection of GAN-generated face images. The generalization ability of the model has also attracted much attention: Wang et al. pointed out that a standard image classifier trained on only one specific CNN generator could generalize to unseen architectures, datasets, and training methods (including the recently released StyleGAN2) [32]. In [33], Liu et al. found that the noise pattern of real images exhibits similar characteristics in the frequency domain, while the generated images are far different. The authors proposed a method by checking whether an image follows the noise patterns of authentic images. This method is proven to be effective in the detection of a wide range of generative models. Ju et al. emphasized the importance of combining information from the whole image and informative patches in improving the generalization ability of AI-synthesized image detection. The authors designed a two-branch model to combine global spatial information and informative local features for the task [34]. In addition, some methods based on spatial information [35–38], Benford's law [39], and frequency information [39–41], are also used to detect GAN images.

In summary, the above mentioned works can achieve promising results under specific scenarios. However, less attention has been paid to the detection of images generated by text-to-image diffusion models. Therefore, it is of great significance to propose a novel algorithm to tackle this issue. In this paper, a novel algorithm is proposed to detect fake images generated by text-to-image diffusion models. The main contributions of this work are stated as follows:

(1) We study the new problem of differentiating fake images generated by text-to-image diffusion models from real ones by leveraging multi-level feature representation learning. (2) A hybrid neural network is constructed by designing attention-guided feature extraction (AGFE)

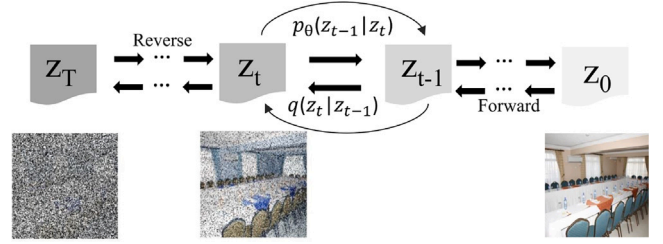


Fig. 1. Illustration of forward and reverse diffusion process.

and vision transformers (ViTs)-based feature extraction (ViTFE) modules for multi-level feature representation learning. The combination of the strengths of CNNs and ViTs facilitates the representation of trace information. (3) Scenarios with different multimodal generative models are considered to evaluate the effectiveness and robustness of the proposed algorithm comprehensively.

2. The acquisition process of DM images

The acquisition of NP images first requires capturing the image, which involves factors such as setting up the shot, focusing, exposure, and taking the shot. Once the image has been captured, it may be processed in various ways to enhance its quality. After that, the processed image is typically stored in a digital format, such as a JPEG or PNG file. Different from the NP image acquisition process, the diffusion model generates images from noise within finite transitions. During training, the input data is gradually perturbed over several steps by adding Gaussian noise. The reverse diffusion stage recovers the original data from the diffused (noisy) data by learning to reverse the diffusion process gradually [42].

As can be seen from Fig. 1, in the forward procedure, Gaussian noise is added to data in each step until the images are destroyed. Given a data distribution $z_0 \sim q(z_0)$, the next distribution z_T and the noised image at arbitrary step t can be generated according to the Markovian process.

The reverse diffusion process reverses the above process and sample from $q(z_{t-1}|z_t)$ to recreate the true sample from a Gaussian noise input $z_T \sim \mathcal{N}(0, \mathbb{I})$. The steps can be formulated as follows:

$$p(z_{t-1}|z_t) := \mathcal{N}(z_{t-1}; \mu(z_t, t), \Sigma(z_t, t)) \quad (1)$$

where $\mu(z_t, t)$ and $\Sigma(z_t, t)$ denote the mean and the covariance respectively. In fact, the reverse procedure starts from z_T , and the model generates $p(z_0)$ which follows the true data distribution $q(z_0)$ by optimizing the following objective function in this iterative process:

$$E_{t \sim [1, T], z_0 \sim q(z_0), \epsilon \sim \mathcal{N}(0, \mathbb{I})} \lambda(t) \left\| \epsilon(z_t, t)^2 \right\| \quad (2)$$

By performing different sampling methods, the model can generate high-quality, realistic images that look like real photographs. According to the above analysis, the inherent traces in the DM images mainly include disturbance traces of noise and compression traces: $\zeta_{DM} = \zeta_{ns} + \zeta_{ic}$ where ζ_{ns} denotes the disturbance traces caused by noise removal operation, and ζ_{ic} is the lossy compression traces of the DM images, which can be formulated as $\zeta_{ic} = [IT(\mathbf{D} \times \mathbf{QF})] - IT(\mathbf{D} \times \mathbf{QF})$, where $[\cdot]$ is the rounding and truncation operations; $IT(\cdot)$ denotes the inverse discrete cosine transform. $\mathbf{QF}$ and $\mathbf{D}$ are the quantization parameter and the coefficients, respectively.

3. The proposed approach

In this work, we proposed a novel approach by exploiting multi-level feature representation learning to expose fake images generated by text-to-image diffusion models.

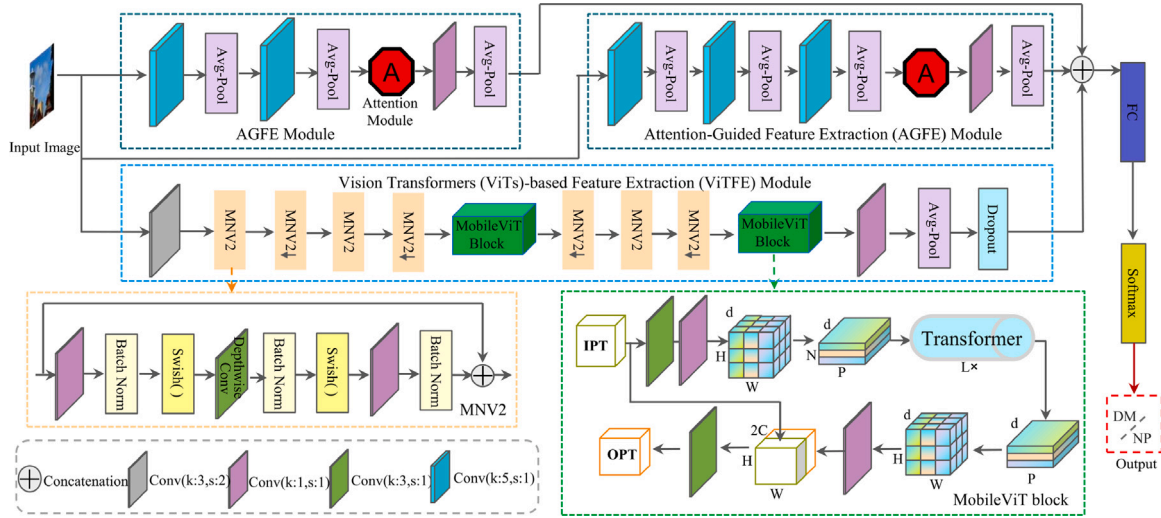


Fig. 2. Overall framework of our proposed approach.

3.1. Attention-guided feature extraction (AGFE) module

Generating an image will inevitably introduce different traces compared with those of real image, we designed two sets of AGFE modules to explore such trace difference information. As shown in Fig. 2, the modules contain three and four sets of convolution and pooling layers, respectively, and each set of modules is equipped with an attention mechanism. The convolution layers before attention module use 5×5 convolution kernel, and the 1×1 kernel is adopted by the subsequent convolution layer.

To refine intermediate features, we apply an attention mechanism in AGFE module, which is flexible and capable of capturing long-range feature interactions and boosting the representation capability [43]. The attention module includes a channel attention submodule $\text{CA}(\cdot)$ and a spatial attention submodule $\text{SA}(\cdot)$, which are connected in sequential order. The input feature maps of $\text{CA}(\cdot)$ is a series of single feature maps, which can be represented by $\text{CA}_{in} = [F_{ca}^1, F_{ca}^2, \dots, F_{ca}^k, \dots, F_{ca}^K]$, k and K in the formula denote the index and the number of feature maps, respectively. These feature maps are fed into a global average-pooling layer to obtain K feature maps of size 1×1 , the k th of which can be expressed as:

$$\text{FV}^k = \frac{1}{A \times B} \sum_{a=1}^A \sum_{b=1}^B f v^k(a, b) \quad (3)$$

where $A \times B$ is the size of the input feature map, $f v^k(a, b)$ is the value at position (a, b) of F_{ca}^k . After that, the channel-wise statistics are downsampled with a scaling ratio r by using a 1×1 convolution layer. Further, another convolutional layer is used to perform the $r \times \frac{K}{r}$ upscaling operation. We fed the upscaled feature map into a sigmoid gate to obtain the channel attention map, which is multiplied by the original feature maps to obtain the final refined feature maps. The procedure can be formulated as follows:

$$\begin{aligned} \text{CA}_{out} &= \text{CM}(\text{CA}_{in}) \otimes \text{CA}_{in} \\ &= \text{Sig}(\text{CL} \uparrow (\text{CL} \downarrow (\text{AP}(\text{CA}_{in})))) \otimes \text{CA}_{in} \end{aligned} \quad (4)$$

where CA_{out} is the output of the channel attention submodule. $\text{CM}(\cdot)$ denotes channel attention mapping operation. $\text{Sig}(\cdot)$ is the sigmoid function. $\text{AP}(\cdot)$ is the average-pooling operation. $\text{CL} \downarrow (\cdot)$ and $\text{CL} \uparrow (\cdot)$ are the convolutional layers used for downscaling and upscaling, respectively.

To capture the spatial relationships of different feature maps, CA_{out} is sent to the spatial attention submodule $\text{SA}(\cdot)$, which generates a spatial attention map by utilizing the inter-spatial relationship of features [43]. The procedure of $\text{SA}(\cdot)$ can be formulated as $\text{SA}_{out} =$

$\text{SM}(\text{CA}_{out}) \otimes \text{CA}_{out}$, where $\text{SM}(\cdot)$ is a 2-dimensional spatial attention mapping operation through average-pooling and max-pooling operations along the channel axis. Then, feature concatenation and additional convolutional operations are conducted on these two features. A sigmoid gate is applied to obtain the $\text{SM}(\cdot)$. The procedure can be formulated as follows:

$$\begin{aligned} \text{SA}_{out} &= \text{SM}(\text{CA}_{out}) \otimes \text{CA}_{out} \\ &= \text{Sig}(\text{CL} \uparrow (\text{AP}(\text{CA}_{out}), \text{MP}(\text{CA}_{out}))) \otimes \text{CA}_{out} \end{aligned} \quad (5)$$

The outputs of the two sets of AGFE modules will be concatenated and fed into the fully connected (FC) and the Softmax layers as part of the classification features.

3.2. Vision transformers (ViTs)-based feature extraction (ViTFE) module

Since global traces can provide reliable information for detecting fake images generated by text-to-image diffusion models, we designed a Vision transformers (ViTs)-based feature extraction (ViTFE) module to learn global representations.

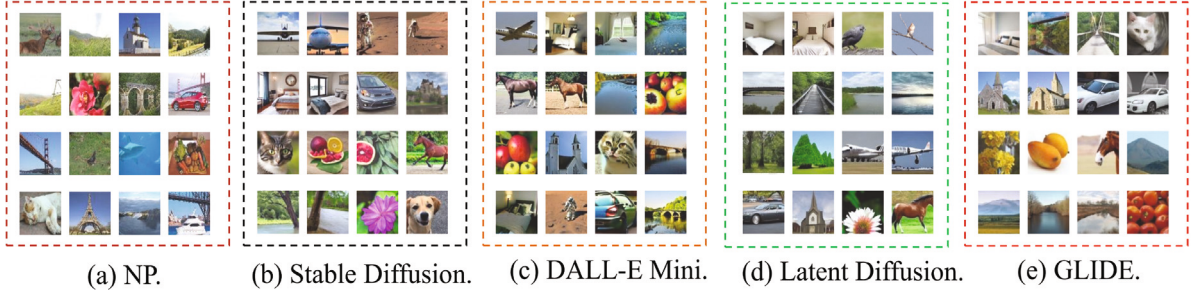
As shown in Fig. 2, the module contains sequential MobileNetV2 block (MN2) and MobileViT blocks for feature refinement. Before the first MN2 block, a standard 3×3 convolution layer is added in this branch. MN2 block consists of several layers, including a depthwise separable convolution layer, a linear bottleneck layer, and a skip connection. The depthwise separable convolution layer is a lightweight alternative to standard convolutional layers, which factorizes the convolution operation into two separate operations: depthwise convolution and pointwise convolution. The linear bottleneck layer is added to increase the representational power of the model. It consists of a 1×1 pointwise convolution layer followed by a nonlinear activation function in which the Swish [44] is used, and another 1×1 pointwise convolution layer. This layer reduces the number of channels in the feature maps and introduces nonlinearity. Finally, the skip connection is added to enable the gradient flow during backpropagation and improve the training of the network. The skip connection enables the input directly forward to the output, which allows the model to learn residual mappings and reduce the likelihood of vanishing gradients. MN2 $\downarrow$ represents a down-sampling MN2 block.

MobileViT block replaces local processing in convolutions with global processing using transformers, which allows the MobileViT block to have CNN- and ViT-like properties, making it learns better representations with fewer parameters and simple training recipes [45]. The feature refining process in the MobileViT block is shown in the green dotted box at the lower right corner of Fig. 2. The input tensor $\text{IPT} \in$

Table 1

Comparison with existing methods on fake image detection. The methods were trained, validated and tested on DM images at a ratio of 5:1:1.

Methods	No compression and noise addition			JPEG compression (QF=85)			JPEG compression (QF=75)			Gaussian white noise (Mean=0.02)		
	ACC (%)	TPR (%)	TNR (%)	ACC (%)	TPR (%)	TNR (%)	ACC (%)	TPR (%)	TNR (%)	ACC (%)	TPR (%)	TNR (%)
Proposed	97.00	95.35	98.65	93.25	93.08	93.42	88.50	86.75	90.25	90.15	90.55	89.75
Rah2017 [22]	58.25	55.40	61.10	55.25	53.75	56.75	53.05	55.82	50.28	54.25	51.85	56.65
Qu2018 [24]	75.38	75.05	75.71	70.25	68.60	71.90	64.28	62.40	66.16	61.20	59.82	62.58
Yao2022[46]	94.25	92.88	95.62	90.73	88.58	92.88	86.87	85.68	88.06	85.48	85.25	85.71
Ga2022 [47]	82.18	80.85	83.51	75.35	73.26	77.44	67.09	66.20	67.98	68.35	66.40	70.30
Mo2018 [29]	85.35	82.20	88.50	82.35	79.80	84.90	77.25	74.43	80.07	74.32	71.35	77.29
Liu2020 [36]	92.88	90.25	95.51	90.32	89.85	90.79	86.28	83.70	88.86	86.50	86.72	86.28

**Fig. 3.** Examples of NP and different types of DM images.

$\mathbb{R}^{H \times W \times C}$ is first fed to a CNN module containing a 3×3 and a 1×1 convolution layers to obtain the local representation of the feature. $H \times W \times C$ represents the size of **IPT**. The output high level feature $\lambda \in \mathbb{R}^{H \times W \times d}$ is then unfolded into N non-overlapping patches, which can be denoted as $\lambda_{fp} \in \mathbb{R}^{P \times N \times d}$, where $P = hw$, $N = H/h \times W/w$. After obtaining these patches, a transformer module is adopted to capture the relationships between different patches. Then, we perform a fold operations on the output λ'_{fp} to obtain a $H \times W \times C$ feature map, i.e., $\lambda' \in \mathbb{R}^{H \times W \times d}$, which is further fed into a convolutional layer with a kernel size of 1×1 . The produced feature map is concatenated with **IPT**, which is also fed into a convolutional layer with a kernel size of 3×3 to obtain the output tensor **OPT**.

After the last MobileViT block, a global average pooling layer is inserted, and a dropout layer is also added at the end of the module to avoid overfitting. The output of the ViTFE module is also concatenated and fed into the FC and the Softmax layers for classification. The classification procedure can be represented as follows:

$$\text{output} = SF(FC(CT(\mathbf{FM}_{\text{AGFE}}, \mathbf{FM}_{\text{ViTFE}}))) \quad (6)$$

where $\mathbf{FM}_{\text{AGFE}}, \mathbf{FM}_{\text{ViTFE}}$ are the outputs of AGFE and ViTFE modules. $CT(\cdot)$, $FC(\cdot)$, and $SF(\cdot)$ denote the feature concatenation, fully connection and the softmax operations, respectively.

4. Experiments and discussion

In this section, we present experiments and discussions to evaluate the performance of our approach. First, the experimental setup is given, including the dataset, parameters, and evaluation criterion. Then, we compare our algorithm with several well-known neural networks and methods.

4.1. Experimental setup

In this work, we conducted experiments on 40000 NP images and 40000 DM images generated by several state-of-the-art pre-trained generative models, including Stable Diffusion [3], DALL-E Mini [48], Latent Diffusion [49], and GLIDE [50] models. We used each type of DM model to generate 10,000 images with different content. The keywords of prompts used to generate DM images include *aircraft*,

astronaut, bedroom, bird, bridge, car, cat, church, fruit, horse, river, dog, flower, castle. Some examples are shown in Fig. 3. It can be observed that it is difficult to distinguish fake images from real ones with the naked eye, and it is also impossible to differentiate images generated by different models. The NP images with different contents were downloaded from the Columbia [51], RAISE [52], LSUN, and BOSSbase v1.01 [53] datasets. We cropped these images to 256×256 due to their inconsistent sizes, and defined CG and NP images as positive and negative samples, respectively. The ratio of training, test, and validation sets is set to 5:1:1. Note that no lossy compression trace is introduced to these images in the cropping process.

In terms of the construction of network architecture, Pytorch-1.11.0, with CUDA-11.3 and Torchvision-0.12.0 packages, is selected in this work. We use the Adam optimizer with a batch size of 32, and the initial learning rate is set to 0.00001 for training. h and w in MobileViT are fixed at 2. The random horizontal and vertical flip probabilities are fixed at 0.3. We train our framework with cross-entropy loss to obtain the detection results. Accuracy is used to measure the detection performance, which can be expressed as $ACC = \frac{TP+TN}{P+N} \times 100\%$, where TP and TN denote the number of true positive and true negative samples, respectively. P and N represent the number of positive and negative samples. We also calculate the True Positive Rate (TPR) and True Negative Rate (TNR) as supplementary metrics, i.e., $TPR = \frac{TP}{TP+FN} \times 100\%$ and $TNR = \frac{TN}{TN+FP} \times 100\%$.

4.2. Performance evaluation on fake image detection

In this section, we evaluate the performance of the proposed approach on the detection of fake images generated by text-to-image diffusion models. We compare this method with a series of algorithms, including Rah2017 [22], Qu2018 [24], Yao2022 [46], Ga2022 [47], Mo2018 [29], and Liu2020 [36], among which Rah2017 [22], Qu2018 [24], and Yao2022 [46], can achieve promising performance on computer-generated image detection. The other methods have achieved accuracies higher than 90% in GAN image detection. Note that, we constructed the neural networks by following the specified structure and parameters in the respective papers, and trained it on the training set. The trained model was used for testing. By training and testing the models in different scenarios, results are obtained as shown in Fig. 4, Table 1, and Table 2.

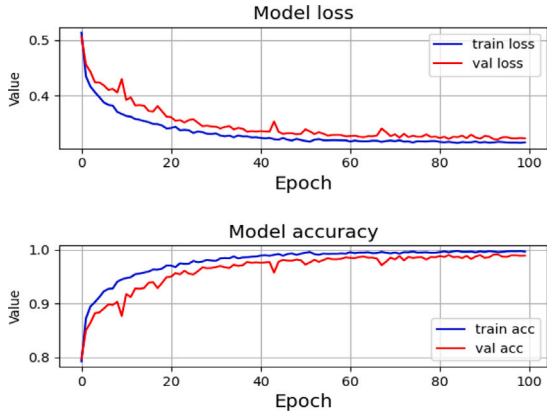


Fig. 4. The visualization of the training and validation loss/accuracy of the proposed network.

Fig. 4 shows that the training loss and validation loss of our proposed neural network decrease significantly in the first 40 epochs, and they plateau after around 70 epochs. The trend of accuracy rates is opposite to that of losses, which increases significantly in the first 40 epochs and stabilizes after 70 epochs. It can be observed from Table 1 that our approach can achieve consistently higher accuracies in most cases. For the performance on uncompressed images, the proposed approach achieves the average accuracy rate of 97.00%, which is 38.75%, 21.62%, 2.75%, 14.82%, 11.65%, and 4.12% higher than those of the comparison approaches, respectively. As the QF decreases, the accuracy rate gradually drops, mainly because the smaller the QF, the more significant the information loss caused by image compression, making it more difficult to perceive the generated trace. Anyway, our approach still achieves better results on the compressed images. The average accuracies of our algorithm, Rah2017 [22], Qu2018 [24], Yao2022 [46], Ga2022 [47], Mo2018 [29], and Liu2020 [36] respectively reach 93.25%, 55.25%, 70.25%, 90.73%, 75.35%, 82.35%, and 90.32% when QF=85. Our method achieves an accuracy of 90.15% when Gaussian white noise (Mean=0.02) was added. These results imply that our algorithm could achieve better performance than existing algorithms in fake image detection. This phenomenon is understandable because AGFE and ViTFE modules are inserted in our neural network, which facilitates the learning of the deep representations of the trace information. Results in Table 2 also show promising performance when the method was trained on latent diffusion images, and tested on latent diffusion or stable diffusion images. It is because these architectures share similar artifacts. Different artifacts lead to performance degradation on other types of DM images.

4.3. Ablation study

In order to verify that the introduced blocks are indeed making contribution to the detection performance, an ablation study is conducted on our method. Neural networks without (wo) AGFE, wo ViTFE, wo channel/spatial attention, with attention modules connected in different ways are compared in this experiment. Input images wo horizontal/vertical flipping are also used for experimental verification. Table 3 demonstrates the experimental results on our dataset.

As can be seen from Table 3, the model equipped with AGFE and ViTFE modules clearly outperforms those without the modules. The ACC decreases by 1.85% when the AGFE module is not used, and the value drops by 2.13% when the ViTFE module is removed. The phenomenon clearly indicates that these modules can enhance the learning ability of detailed clues. The performance of the method also decreases when no flipping operation is applied to the input images. It can also be observed that the attention scheme has an impact on the

Table 2

Comparison on DM image detection. The method was trained on latent diffusion images, and tested on different types of DM images.

Tested DM images	Trained on latent diffusion		
	ACC (%)	TPR (%)	TNR (%)
Latent Diffusion	99.32	99.08	99.56
Stable Diffusion	98.87	99.40	98.34
DALL-E Mini	68.12	65.05	71.19
GLIDE	62.88	60.05	65.71

Table 3

Performance of the model with different settings.

Settings	Criteria(%)		
	ACC	TPR	TNR
Proposed	97.00	95.35	98.65
wo AGFE	95.15	94.38	95.92
wo ViTFE	94.87	95.08	94.66
wo horizontal flip	96.60	95.25	97.95
wo vertical flip	96.37	95.05	97.69
wo flip	95.12	95.00	95.24
spatial+channel	96.14	94.88	97.40
wo channel attention	96.05	94.76	97.34
wo spatial attention	95.77	94.83	96.71
wo attention	95.08	94.32	95.84

Table 4

Comparison with existing methods on Cross-dataset image detection.

Methods	ACC (%)	TPR (%)	TNR (%)
Proposed	96.57	95.25	97.89
Rah2017 [22]	56.25	55.15	57.35
Qu2018 [24]	70.80	72.38	69.22
Yao2022[46]	92.27	93.06	91.48
Ga2022 [47]	80.37	78.85	81.89
Mo2018 [29]	78.25	75.33	81.17
Liu2020 [36]	87.32	88.50	86.14

performance. The detection accuracy reaches 96.05% when the channel attention is removed, and this value becomes 95.77% in the absence of the spatial attention module. Therefore, different settings would affect the performance in detecting DM images, and the results verify the effectiveness of the proposed network architecture.

4.4. Cross-modal and cross-dataset image detection

In the experiment introduced above, intra-class evaluation is considered. However, the detection approach may suffer performance degradation if images from unknown sources are used for testing. We therefore evaluate the performance of the approach on cross-modal fake image detection. The following five scenarios are considered in this experiment: (a)(b) Training on DM images, and testing on TCG/GAN images. (c)(d) Training on GAN/TCG images, and testing on DM images. (e) Training on DM images in our dataset, and testing on DM images in UNINA dataset [54]. The TCG image set was created by collecting CG video or game screenshots (Forza Horizon, GANTZ, God of War, Red Dead Redemption, and Playerunknown's Battlegrounds) from Kaggle and Google websites. QF is selected from {90, 80, 70} to evaluate the robustness. We utilized pretrained StyleGAN2 [55], ProGAN [56], and BigGAN [57] models for GAN image generation. The DM images in the test set of UNINA dataset are used for evaluation. By conducting training and testing on different image sets, the results can be obtained, as shown in Fig. 5 and Table 4.

It can be observed from Fig. 5 and Table 4 that the proposed approach performs better than the comparison methods in detecting unseen images, no matter how the training-testing combination

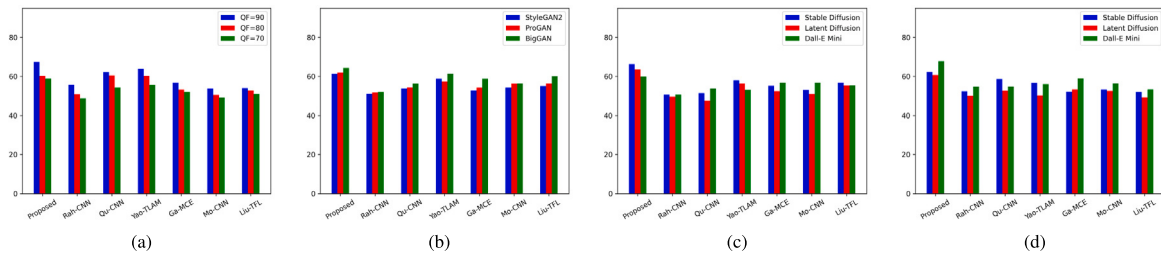


Fig. 5. Performance evaluation on cross-modal image detection (%). (a) Training on DM images, and testing on TCG images. (b) Training on DM images, and testing on GAN images. (c) Training on GAN images, and testing on DM images. (d) Training on TCG images, and testing on DM images.

changes. When the model is trained on DM images and tested on TCG images, our approach achieves average accuracies of 67.40%, which is 11.62%, 5.15%, 3.52%, 10.56%, 13.54%, and 13.35% higher than those of the comparison approaches, respectively. Our method achieves an average accuracy of 96.57% on the UNINA dataset, which is also better than the comparison algorithms. It is because multi-level feature representation learning is leveraged for the task, and generated traces can be captured during the learning process. In summary, the proposed approach outperforms the comparison algorithms in this cross-modal detection scenario.

5. Conclusion

In this work, a novel approach is proposed for exposing fake images generated by text-to-image diffusion models. We first analyzed the acquisition process of DM images and the discriminative traces left in the images. Then, a hybrid neural network was constructed by designing AGFE and ViTFE modules for multi-level feature representation learning. The combination of the strengths of CNNs and ViTs facilitates the representation of trace information. By conducting extensive experiments on different types of generated images, the experimental results demonstrate that the proposed approach can achieve better performance and has a much more robust detection capability than existing methods.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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